

Introduction

Despite consistent efforts to eliminate polio, significant gaps remain in ensuring equitable access to vaccines in Pakistan. These gaps are not merely logistical but are deeply rooted in the structure of the health system and broader societal challenges. Many rural areas lack trained healthcare personnel and reliable cold-chain systems, which are crucial for storing and transporting vaccines safely. The polio vaccine requires a specific temperature to remain both safe and effective. In conflict-affected regions, frequent disruptions to vaccination campaigns due to violence, curfews, or local unrest create prolonged periods when children remain unvaccinated (Ghafoor, 2016). These challenges are compounded by low levels of community engagement, where public health messages often fail to resonate with local beliefs or address the specific concerns of the population. Even when vaccines are available, uptake remains low. The failure to consistently reach these communities undermines not only the national polio eradication strategy but also international commitments to health equity and disease prevention. Addressing this issue requires more than short-term fixes; it demands a coordinated, long-term policy response that strengthens local health capacity and builds trust at the community level.

Description of Policy Problem

Access to the polio vaccine remains a significant challenge in Pakistan, particularly in rural and conflict-affected areas where children are often missed. While national vaccination campaigns have made progress in urban centers, many families in remote or insecure regions still face major obstacles. These include poor infrastructure,

a lack of nearby health facilities, and safety concerns that make it difficult for health workers to reach certain communities. Misinformation and mistrust surrounding the vaccine further discourage families from participating. Pakistan is one of only two countries where wild poliovirus continues to spread, and the risks associated with this extend beyond its borders. Approximately 85% of the world's current polio cases come from Pakistan and Afghanistan. The national rate of fully immunized children in Pakistan is around 76%, with Khyber Pakhtunkhwa at only 42.8% (Khan, 2024). Every missed child not only poses a personal health risk but also represents a setback in the global effort to eradicate polio. Addressing this issue is urgent, not only to protect the health of children in Pakistan but to ensure that global progress toward a polio-free future is not lost.

Policy Option & Policy Barriers

Current policy options that exist to help increase access to the polio vaccine, for community engagement, and to combat misinformation spread of the vaccines include public private partnerships in Pakistan. Approximately 50,000 children are excluded from the government's annual supplementary immunization activities (SIAs) due to vaccine hesitancy. This hesitancy is primarily driven by misconceptions about the vaccine, a lack of trust in healthcare workers, and growing apathy towards polio vaccination efforts, which has been referred to as "polio fatigue" (Khan, 2024). Polio fatigue is when weariness from immunization activities and the need for continued vaccination leads to a decrease in participation. The Pakistan Polio Eradication Programme has implemented the *Naunehal* project, which was distributed through three high-risk union councils in Pakistan. This project delivered immunization and nutrition

services through community mobilization and engagement, which involved raising awareness via health information sessions that focused on child immunization & infant & young child feeding practices (Khan, 2024). A benefit of this project was that the missed opportunities for vaccinations in these areas decreased, as workers returned to the same location for follow-up vaccinations. The Nauhen model optimized staffing and resources, leading to efficiency.

Although the Pakistan Polio Eradication Programme operates as a public-private partnership (PPP) led by the federal government, there is still room to strengthen these partnerships, especially when they are designed with a strong focus on community-based delivery. PPPs can improve service reach and efficiency, but they often fall short in building community trust. In many areas, there is deep-rooted suspicion around foreign involvement in vaccination efforts, which can lead to resistance. Additionally, PPPs tend to prioritize scale and coverage, sometimes overlooking the critical need to protect frontline vaccinators in insecure regions. In contrast, a government-led, community-based approach that actively involves religious leaders and local influencers could help rebuild trust, foster ownership, and reduce vaccine hesitancy. Centering the voices of communities isn't just strategic, it's essential to ensuring long-term success and safety for all involved

To ensure that Sustainable Development Goal 3b, which aims to provide access to affordable essential vaccines, is achieved through local NGO initiatives, community health workers (CHWs) were deployed in rural areas where outbreaks are occurring. This aligns with the World Health Organization's recognition of CHWs as essential to

strengthening the health system and a fundamental component of primary health care. To improve vaccination rates in rural areas of Pakistan, models must focus on strengthening community-based health systems and building trust in health services. CHWs should be trained in health education to address misconceptions and vaccine resistance. The lack of trust in the health system is a major barrier to outreach programs and vaccination efforts. Collaborating with local leaders and religious figures in northern Pakistan could help increase vaccination rates. Faith leaders are highly respected members of the community, and their advice is trusted. A recent example from India demonstrates this approach: during the COVID-19 surge in 2021, a project team enlisted faith leaders to debunk vaccine myths and promote vaccine safety on platforms like WhatsApp. This resulted in an increase of approximately 0.41 million beneficiaries by January 2023 (Soni, 2023). Additionally, during Eid celebrations, India made vaccination a requirement for mosque participation, leading to increased engagement. The article concluded that partnerships between faith-based organizations and vaccination programs were most effective in boosting participation during public health crises (Soni, 2023). In Pakistan, a Muslim-majority country with conservative northern areas, vaccine hesitancy has been exacerbated by concerns over the vaccine's "halal" status (Rahim, 2021). Proposing the involvement of faith-based leaders in community health systems could help address these concerns and target a wider population. Equipping trusted religious leaders with accurate information about the polio vaccine could play a key role in dispelling myths surrounding the trivalent oral polio vaccine.

Several barriers could prevent the proposed solution of strengthening community health workers and faith-based organizations to increase vaccine participation,

especially in regions like Khyber Pakhtunkhwa, where security concerns have escalated. In these areas, attacks on polio vaccinators have led to significant declines in participation. According to Ghafoor, 40 polio workers have been killed, and public health facilities in the region have come under attack by residents. To address this, a national security protocol should be established for polio vaccination programs. This would involve ensuring that vaccination teams are escorted by security personnel and strengthening the commitment of both the government and the military. Furthermore, it's important to implement mandates that protect health workers and form a security task force to gather intelligence on potential threats. Without addressing these security concerns, continued participation from community health workers in these high-risk areas will be compromised.

Stakeholder Priority Matrix

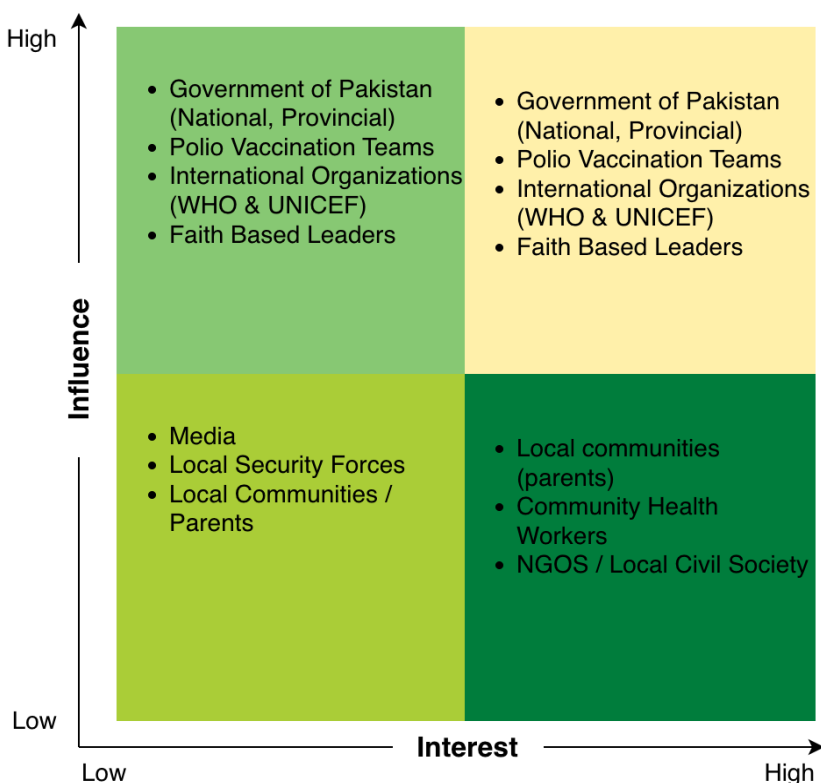


Figure 1. Stakeholder Priority Matrix of Polio Eradication Policy

The stakeholder priority matrix highlights that the Government of Pakistan, both national and provincial, along with polio teams and faith-based leaders, are central to driving progress on eradication efforts, especially when guided by international best practices. Local communities, health workers, and NGOs should be kept informed and involved to build trust and ensure smooth implementation. It's also important to engage media, security forces, and parents, keeping them aware of the challenges and goals behind each initiative. Polio eradication depends on collaboration, clear communication, and the active involvement of those most affected.

Conclusion

In conclusion, significant progress has been made in the fight against polio in Pakistan. Yet, the barriers to vaccination access go far beyond logistical constraints. At the heart of the issue are forgotten communities, health workers who risk their lives daily, and children whose futures depend on protection that should be guaranteed.

These efforts cannot succeed without urgently addressing the critical security risks that continue to endanger vaccinators and discourage families from participating in immunization campaigns. Implementing a national security protocol, ensuring safe working conditions for frontline health workers, and fostering genuine trust within local communities are not just strategic actions, they are moral imperatives. Every missed vaccination is not merely a setback, it's a child left vulnerable, a family at risk, and a threat to decades of global progress. By committing to a government led long-term approach rooted in security, community partnership, and inclusive dialogue, Pakistan has the opportunity to not only eliminate polio but to rebuild trust in public health systems. A polio-free future is within reach, but only if we act with both urgency and empathy.

Appendix

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